Chapter 22: Change of basis Worksheet

Notation reminder: $[\mathbf{v}]_B = C_B(\mathbf{v})$ is the coordinate representation for a vector $\mathbf{v}$ in a vector space V with respect to an ordered basis B. If B and D are ordered bases for V, the transition matrix/change matrix from B to D is denoted $P_{D\leftarrow B}$.

1. Are each of the following statements true or false?

(a) If $P_{D\leftarrow B}$ is a transition matrix from basis B to basis D for vector space V, and $\mathbf{v}$ is a vector in V, then $P_{D\leftarrow B}[\mathbf{v}]_D = [\mathbf{v}]_B$.

(b) If Q is a transition matrix for two ordered bases B and B' for M_{22}, then $\text{rank}(Q) = 4$.

(c) If $P_{C\leftarrow B}$ is the transition matrix from basis B to basis C, and $P_{D\leftarrow C}$ is the transition matrix from C to D, then the transition matrix from D to B is $(P_{C\leftarrow B})^{-1}(P_{D\leftarrow C})^{-1}$.

2. Let $B = (x, x+1)$ and $D = (-2x-3, 3x+4)$ be ordered bases for P_1. You are given the transition matrix from B to D: $P_{D\leftarrow B} = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$.

(a) Use the transition matrix to compute $[p]_D$ for $p = 5x + 8$.

(b) Check your answer to part (a) by writing out the linear combination of p using $[p]_D$.

3. Let $B = (p_1, p_2, p_3)$ be an unknown ordered basis for P_2, and suppose

$P_{B' \leftarrow B} = \begin{bmatrix} 4 & 0 & -1 \\ 2 & 0 & 3 \\ -1 & 1 & 0 \end{bmatrix}$ is the transition matrix from B to $B' = (x^2 + 3, x - 4, x^2 - 4x + 1)$.

Find p_1, p_2, p_3.

4. Consider U, the set of 2×2 symmetric matrices. On a previous worksheet, you showed that $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ and $T = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ are spanning sets for U.

(a) Assuming that S and T span U, show that S and T are bases for U.

(b) Build the transition matrix from T to S (using the given order for each).

(c) Find the transition matrix from S to T (using the given order for each).

(d) Use your previous answer(s) to compute $[M]_T$ for $M = \begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix}$.